

Notes Week 12

a. Analytic

I am basing my implementation on the results from [Soberman et al. \(1997\)](#).
The code below shows my implementation.

```
class CombineEvolve:
    def __init__(self, alpha, beta, delta, gamma, qs, A=1, M_1_i=2):
        self.alpha = alpha # fraction lost from donor as wind
        self.beta = beta # fraction lost from accretor as wind
        self.delta = delta # fraction transferred to CCT
        self.gamma = gamma # measure for CCT distance
        self.epsilon = 1 - self.alpha - self.beta - self.delta
        self.qs = 1 / qs # note that soberman1997 uses q = m_d / m_a
        self.q0 = 1 / qs[0]

        self.A = A # enhancement factor
        self.M_1_i = M_1_i # initial mass of donor

        self.curA = self._comp_curA()
        self.curB = self._comp_curB()
        self.curC = self._comp_curC()

    def _comp_curA(self):
        # equation B8 soberman1997
        return self.A * self.alpha + self.gamma * self.delta

    def _comp_curB(self):
        # equation B9 soberman1997
        return (self.A * self.alpha + self.beta) / (1 - self.epsilon)

    def _comp_curC(self):
        # equation B10 soberman1997
        term_1 = (self.gamma * self.delta * (1 - self.epsilon)) / self.epsilon
        term_2 = (self.A * self.alpha * self.epsilon) / (1 - self.epsilon)
        term_3 = (self.beta) / (self.epsilon * (1 - self.epsilon))
        return term_1 + term_2 + term_3

    @property
    def a_over_a0(self):
        # table 5 soberman1997
        term_1 = (self.qs / self.q0) ** (2 * self.curA - 2)
        term_2 = ((1 + self.qs) / (1 + self.q0)) ** (1 - 2 * self.curB)
        term_3 = ((1 + self.epsilon * self.qs) / (1 + self.epsilon * self.q0)) ** (
            2 * self.curC + 3
        )
        return term_1 * term_2 * term_3

    @property
    def M_tot_over_M_tot0(self):
        # table 5 soberman1997
        term_1 = (1 + self.qs) / (1 + self.q0)
        term_2 = ((1 + self.epsilon * self.qs) / (1 + self.epsilon * self.q0)) ** (-1)
        return term_1 * term_2

    @property
    def P_over_P0(self):
        # table 5 soberman1997
        term_1 = (self.qs / self.q0) ** (3 * self.curA - 3)
        term_2 = ((1 + self.qs) / (1 + self.q0)) ** (1 - 3 * self.curB)
        term_3 = ((1 + self.epsilon * self.qs) / (1 + self.epsilon * self.q0)) ** (
            3 * self.curC + 5
        )
        return term_1 * term_2 * term_3

    @property
    def M_tot(self):
        return self.M_1_i * (1 + 1 / self.q0) * self.M_tot_over_M_tot0

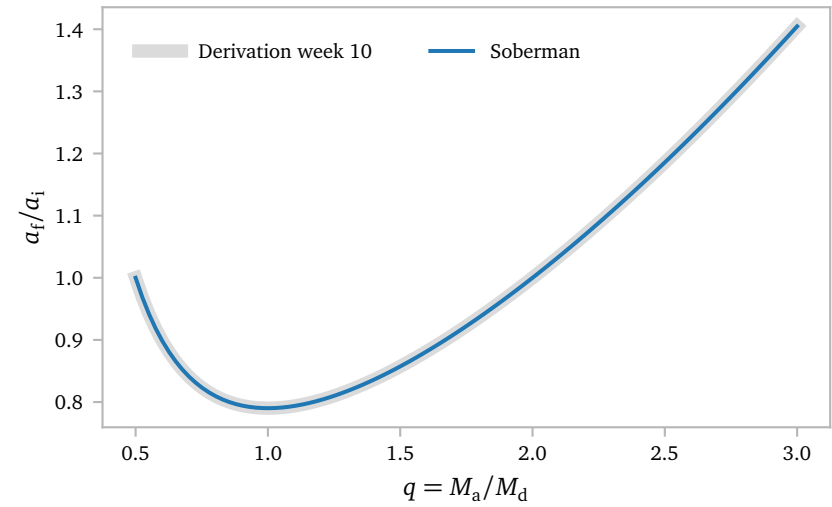
    @property
    def M_d(self):
        return self.M_tot / (1 + 1 / self.qs)

    @property
    def M_a(self):
        return 1 / self.qs * self.M_d
```

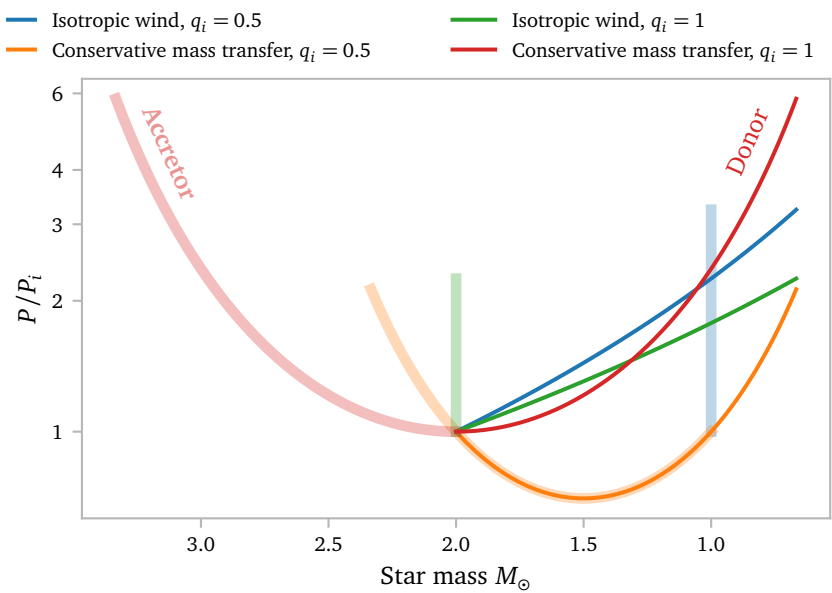
a.1 Exploring Effects of α and β

First, I want to compare my implementation of the equations from [Soberman et al. \(1997\)](#) with my results from week 10, which are specifically derived for the case of conservative mass transfer. This aligns with $\epsilon = 1$ and α, β and $\delta = 0$.

Plugging these values into my [Soberman et al. \(1997\)](#) implementation for a given starting q , we arrive at the following plot:

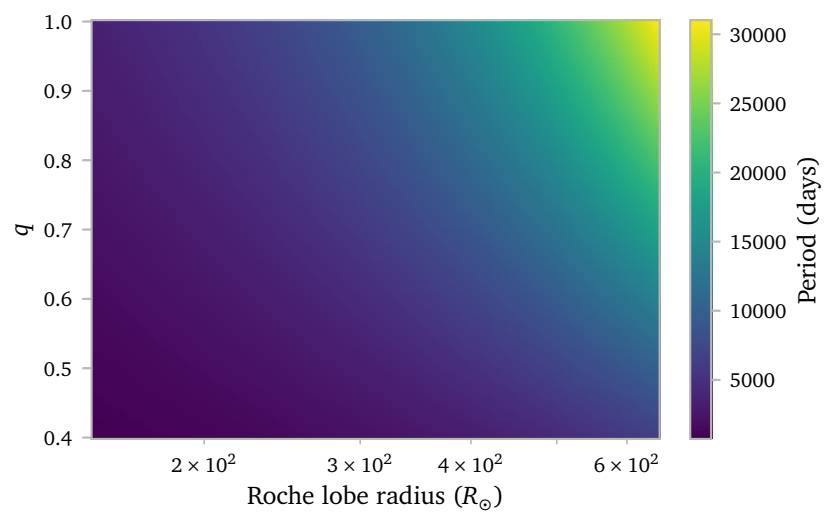


The curves align, and show the expected behaviour. The ratio of a_t/a_i has a minimum at $q = 1$.

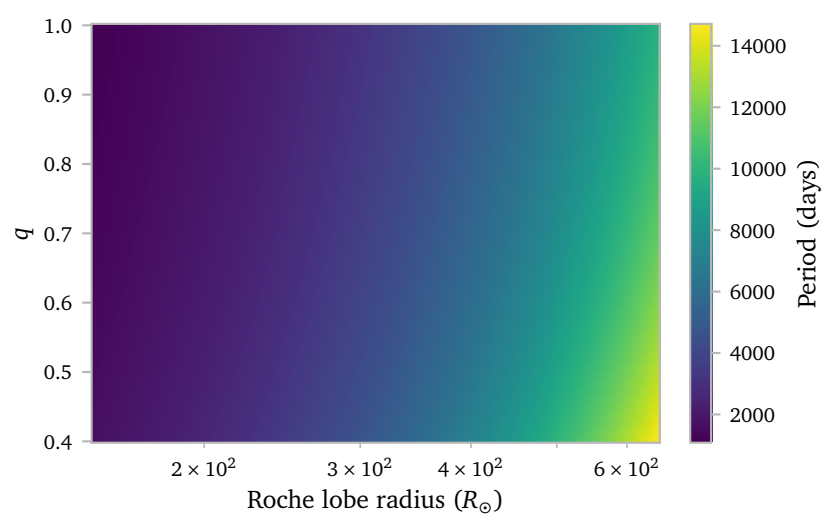


The plot shows four different evolutions. The green and blue lines show a binary system where mass is lost via isotropic winds, while the orange and red lines show a binary system where mass is transferred. In the case of conservative mass transfer, the lines for the donor and accretor are symmetric, which makes sense. In the case of isotropic winds, the accretor does not change, which also makes sense. The effect of the mass ratio $q \equiv M_a/M_d$ is inverted for the two mass loss mechanisms.

We can analytically compute a grid of models, similar to the grid of models that we simulate using MESA.

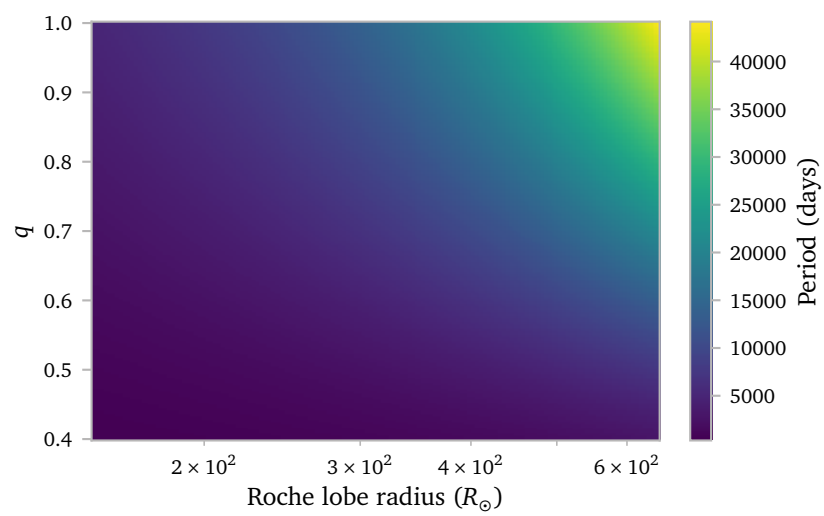


For example, here I show the final orbital period for models that undergo exclusively conservative mass transfer ($\alpha, \beta, \delta = 0$). I make sure the evolution proceeds until the donor star has a mass of $0.6 M_{\odot}$, which is just the core. I think the results make perfect sense.



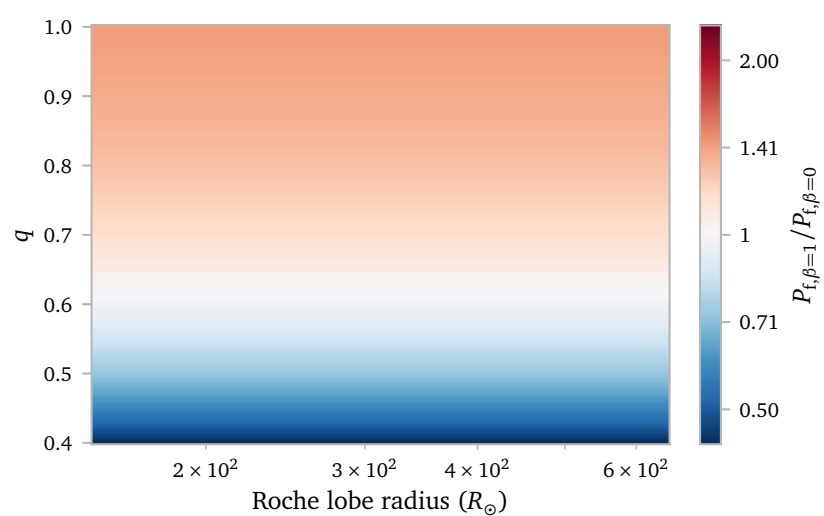
Similarly, we can plot the final orbital periods in the case of only wind mass loss and no accretion, corresponding to $\alpha = 1, \beta = \delta = 0$.

Again, I think these results make sense. We have already seen that wind mass loss has an inverse q dependence on the initial to final period, with lower q generating a greater $\frac{P_f}{P_i}$. Logically, the largest periods are now found for low q and high initial R_{RL} . Overall the periods are smaller, which is also expected.



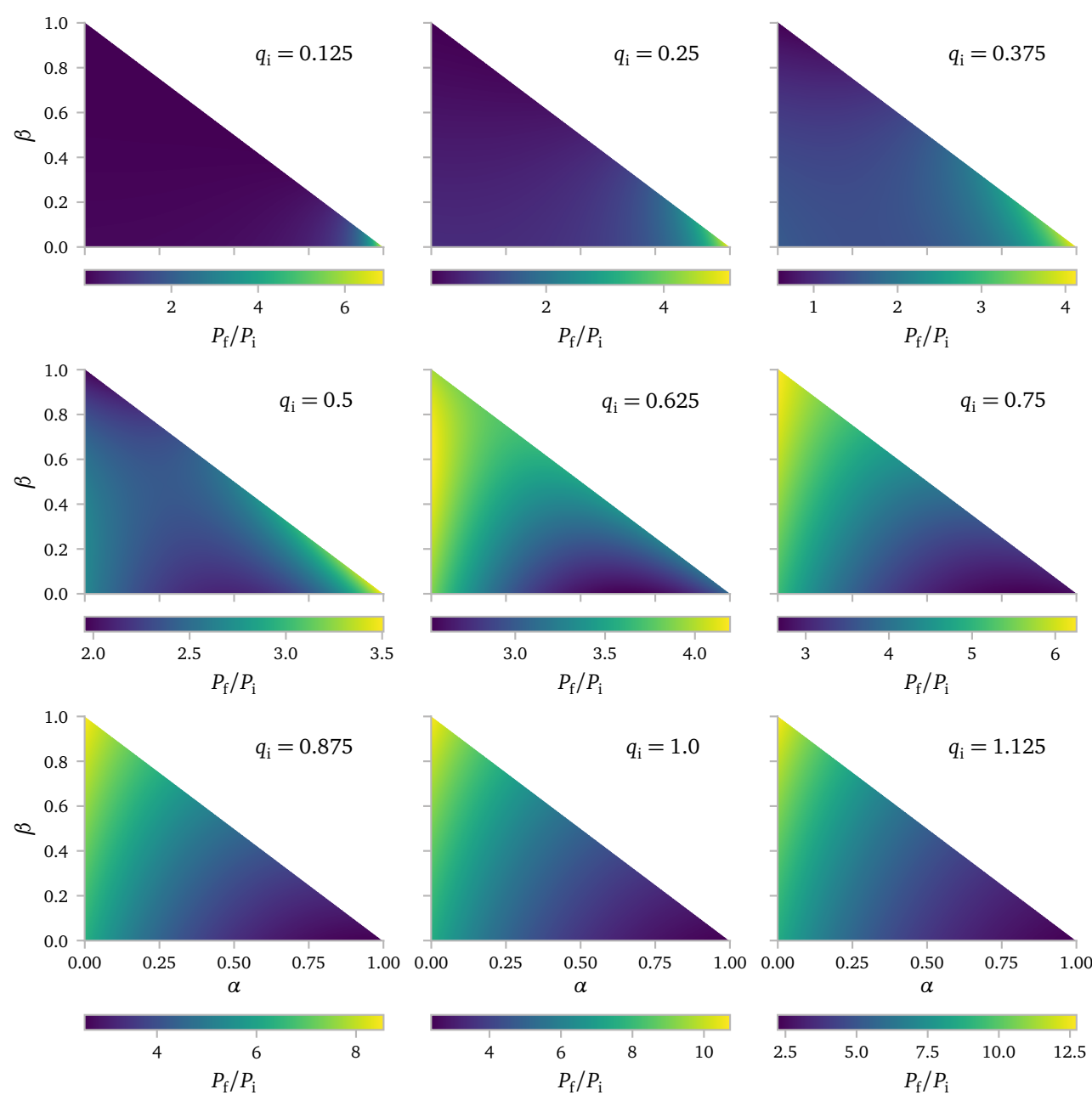
This plot shows the final periods for models with $\alpha = \delta = 0$ and $\beta = 1$, corresponding to the case of pure isotropic re-emission from the accretor.

Since this plot looks very similar to the first plot, I think it is instructive to divide the values from this plot by the values from the first plot, and show these separately.



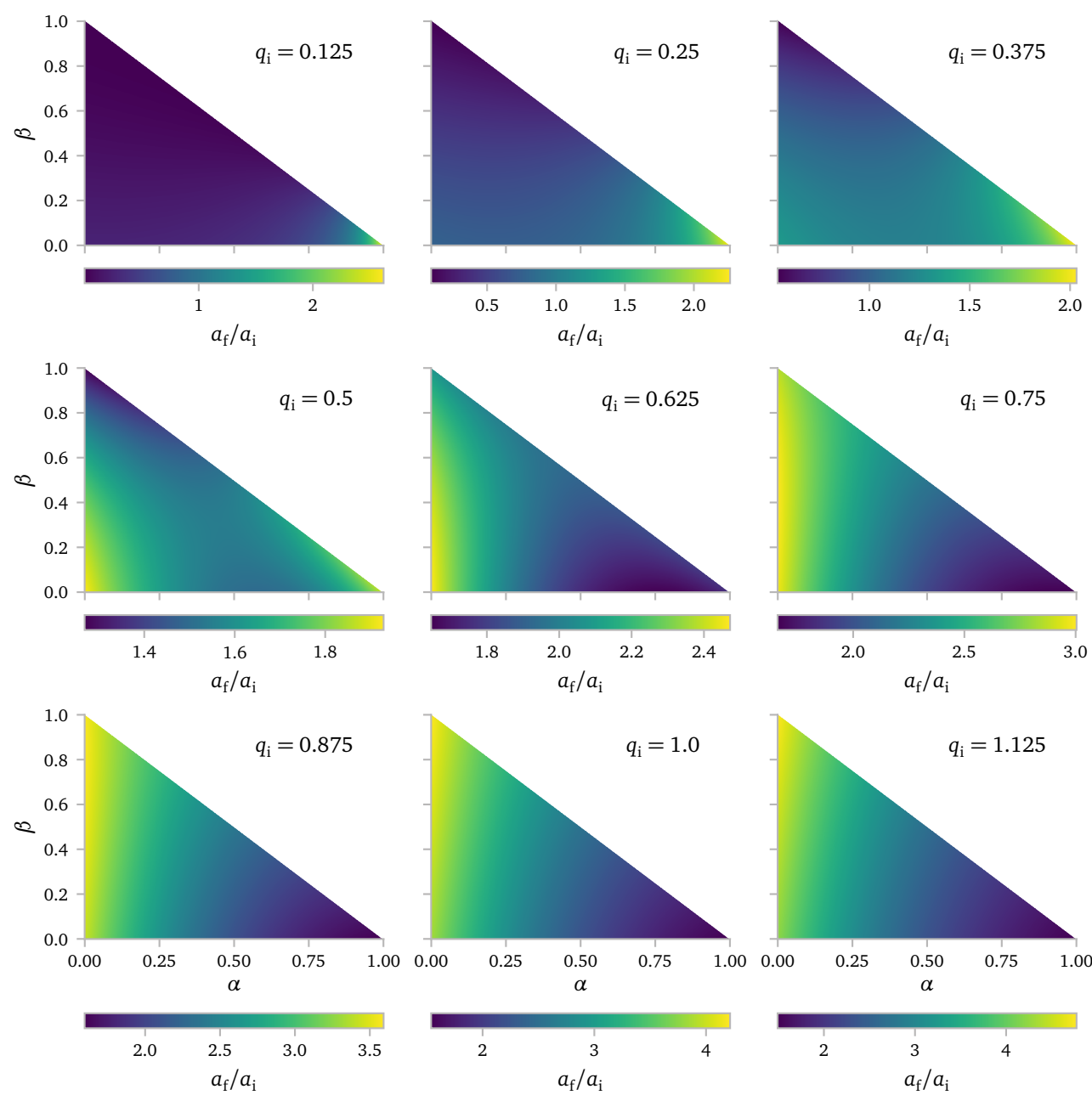
The division is shown here. Clearly, the orbital periods do not get scaled by some constant factor but are quite strongly dependent on the (initial) mass ratio. For relatively low q , the final orbit turns out to be smaller than it would be for conservative mass transfer. For relatively high q , the final orbit is *larger* in the case of isotropic re-emission.

We can also look at the effects of α and β on the orbits. The first plot here shows how period changes with respect to the initial period. I made 9 different plots for increasing initial mass ratios. Every point corresponds with a binary system where the donor initially has a mass of $2 M_{\odot}$ and ends up with a mass of $0.6 M_{\odot}$. The plots are in the shape of triangles as the following condition needs to be true: $\alpha + \beta < 1$.



As can be seen, the initial mass ratio quite significantly alters the effects of α and β . For relatively low initial mass ratios, the orbits will shrink in response to very high β and grow in response to very high α . The inverse is true for mass ratios close to unity.

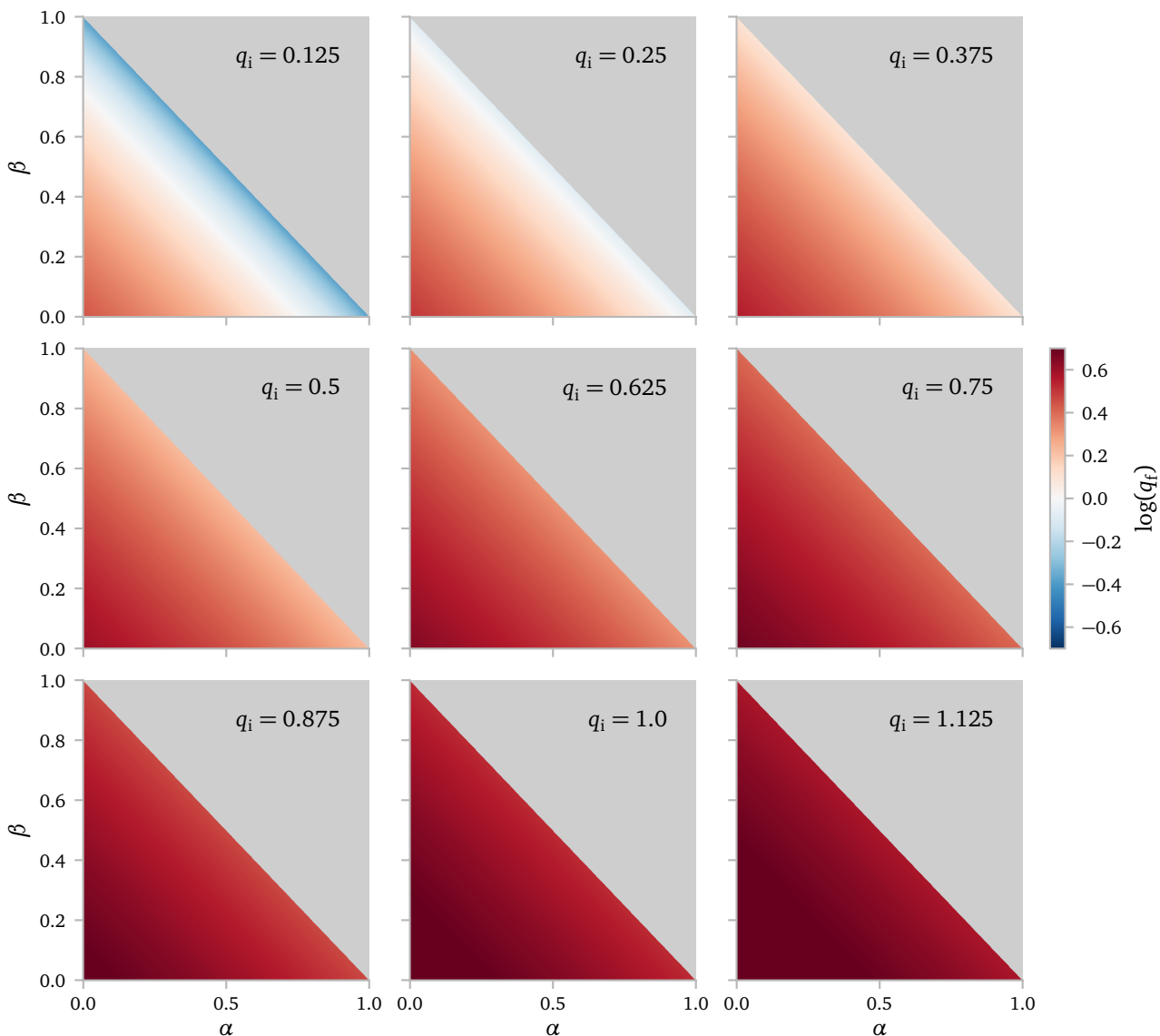
A very similar picture arises when we look at the final separation over the initial separation.



The qualitative behaviour is the same, but the actual differences in ratios are much smaller. Interestingly as well, for low q , the orbits can actually shrink with respect to the initial orbital separation.

These results were not really what I expected. I thought the orbits would *always* shrink when a significant fraction of the mass is lost when accreted onto the accretor. However, this does not seem to be the case. To get a better intuition for the cases where the orbit does shrink with respect to the $\beta = 0$ case, I plot the final and average mass ratios of the binary evolution below. These plots are very structured, which makes sense. The plots also *share* a diverging colorscale. The divergence point lies at $\log q = 0$, which aligns with the case where the stars are equally heavy.

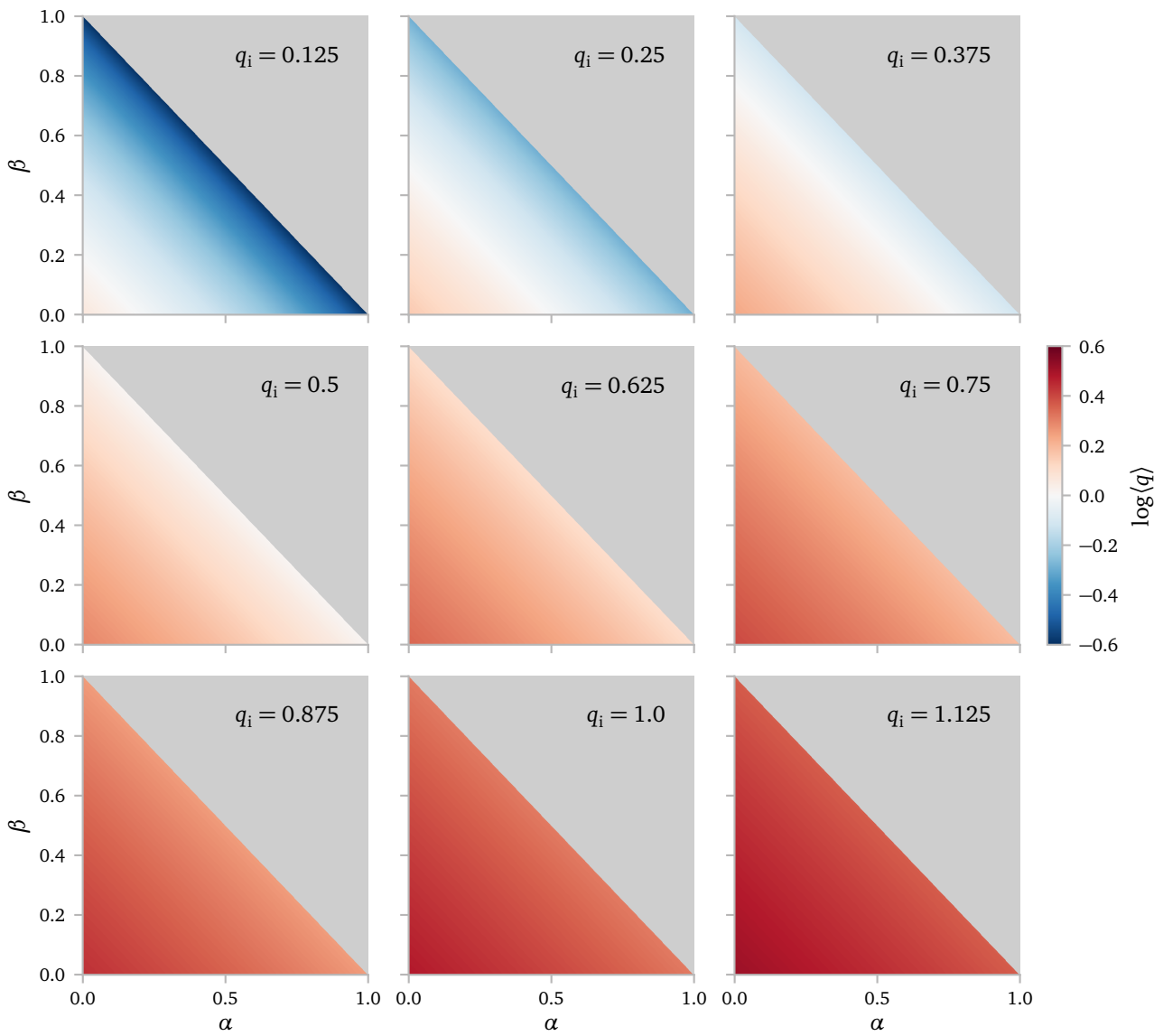
I start by plotting the final mass ratios.



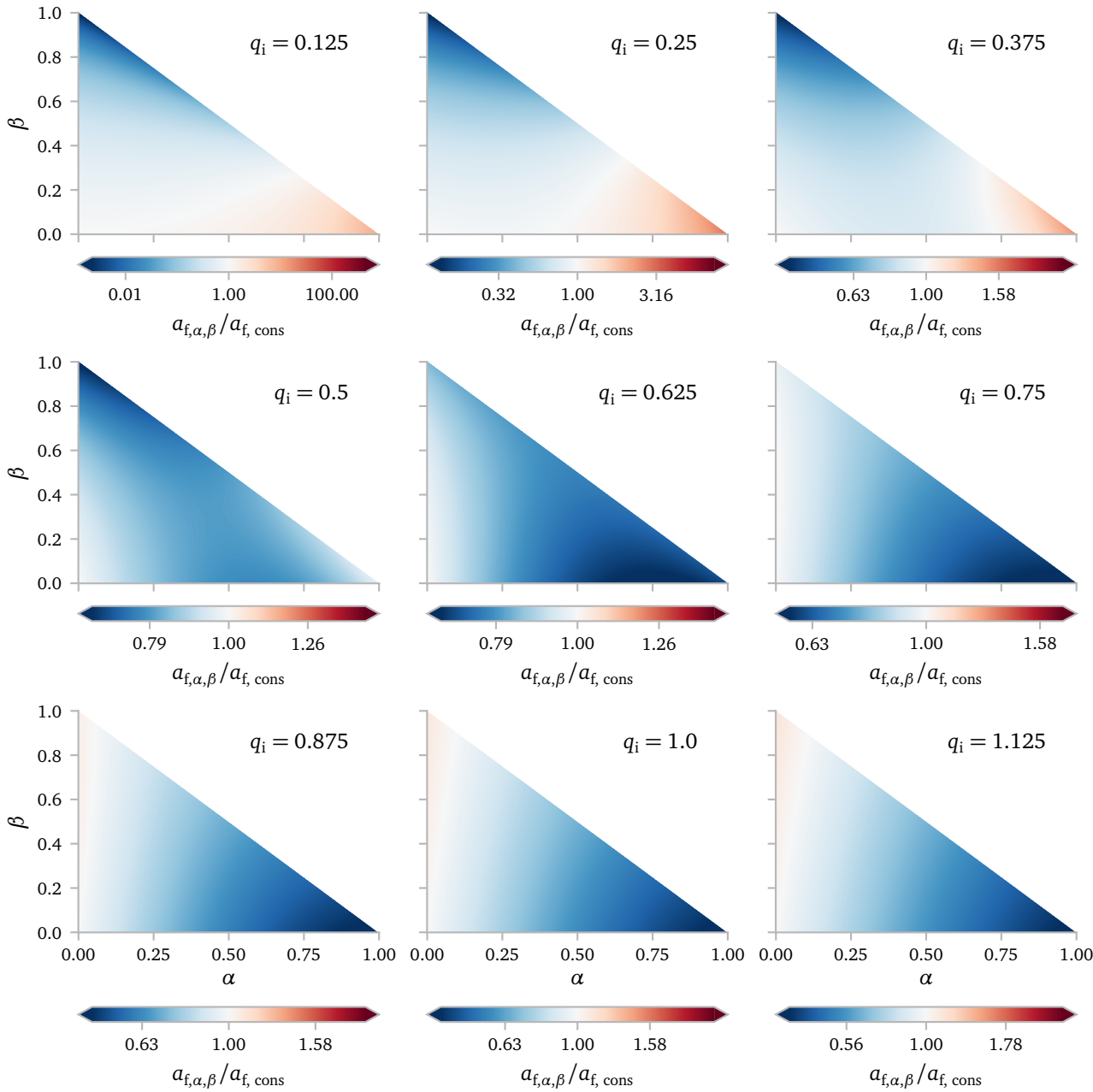
Interestingly,¹ nearly all systems, except for the initially very unequal ones with nearly no accretion, reach a final q that is greater than unity. I *think*, at this point, losing mass at the donor causes the system to lose more angular momentum than losing mass at the accretor.

¹ But after a little bit of thought, very unsurprisingly!

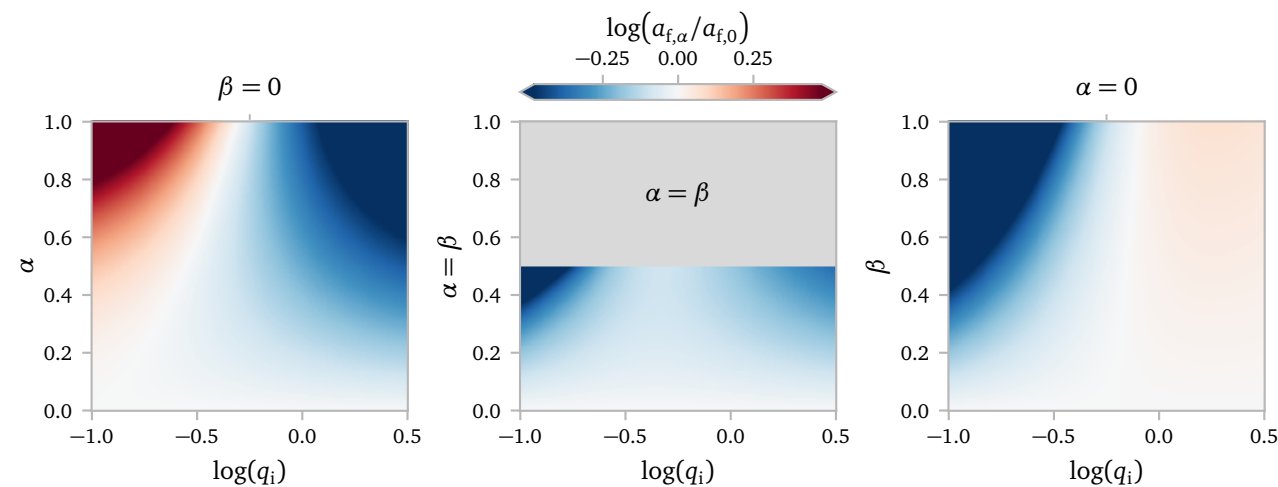
Below I plot the same figure, now for the average q during the evolution.



Now, we see that the systems where $q < 1$ align with the systems where the change in period for $\beta > 0$ is less than for $\beta = 0$.



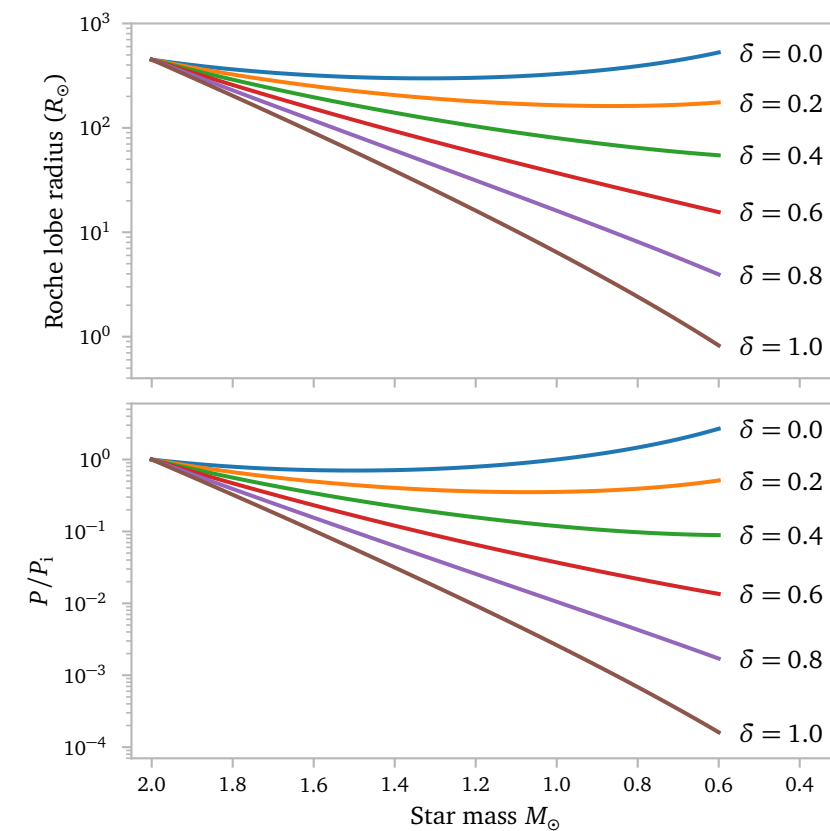
Another way to visualize this, is by tracing cross sections through the plots above for a much finer q spacing. We then obtain the following plots.



a.2 Exploring Effects of a Circumbinary Corotating Toroid

We have seen that winds and isotropic re-emission can have important effects on the properties of the binary orbits. [Soberman et al. \(1997\)](#) however, describe another mechanism that, according to their paper, can be very important for angular momentum loss from an orbit. This is the Circumbinary Corotating Toroid or CCT. It is described by two parameters, δ and γ . The efficiency of mass transport to the CCT is described by β . It is the analogue to α and β for winds and inefficient accretion. The second parameter, γ is a parameter that encodes the specific angular momentum of the ejected material and the specific angular momentum of the binary. In the case of the CCT, this is closely related to the ratio of the CCT and the binary such that as a good first approximation, one can take $\gamma = \sqrt{a_{\text{ring}}/a_{\text{bin}}}$. The difficulty then is in taking a realistic value for a_{ring} . [Soberman et al. \(1997\)](#) propose a modest value of $2.25a_{\text{bin}}$, which corresponds to $\gamma = 1.5$.

With these values, we find the following effects on the orbit for varying strengths of δ and a constant $\gamma = 1.5$:



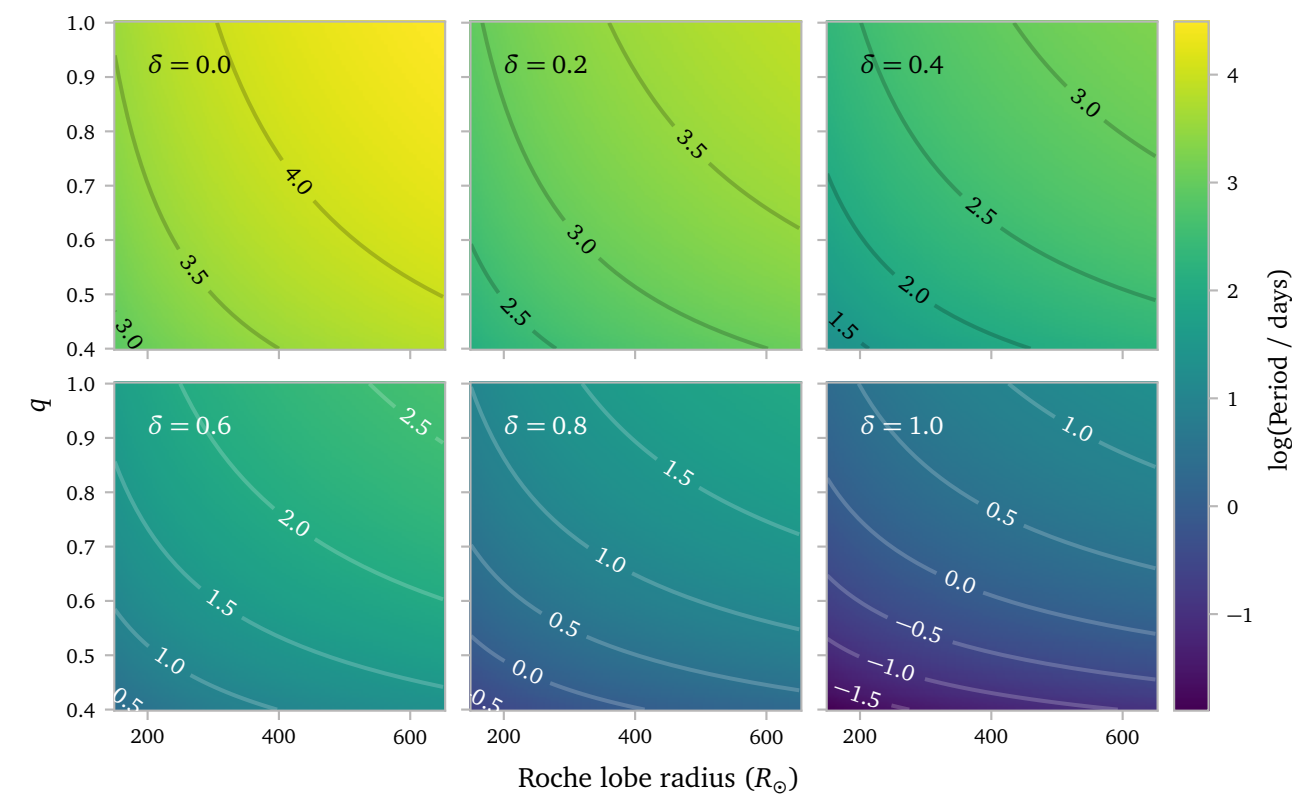
I think these results are quite promising. The orbits are able to shrink *significantly*, even for quite modest δ -values.

To me, it also makes quite a bit of sense for the binaries to develop a CCT. We already see that the donor star gets close to, and sometimes even exceeds the volume equivalent outer Roche lobe radius. Mass that is lost through this point would likely end up in a CCT.

The difficult part in describing this effect is in the determination of the efficiency parameters. Maybe a constant δ value makes some sense because the star will only really lose a lot of mass either through winds at the end of its evolution (by which this mechanism should be unaffected) and through RLOF which only happens during thermal pulses where $R_{\text{star}}/R_{\text{ROL}}$ does not vary that much and is close to order unity throughout the mass transfer episode.

Still, I am not sure *what* δ should be. Maybe this needs to be tuned to observations.

Below, I show the effects of the varying δ for a grid of models with varying Roche lobe radii R_{RL} and mass ratios q . I draw contour lines as well to make it easier to interpret the results. These systems do *not* include any α or β , so this is strictly mass escaping to the CCT where the remainder of the mass is accreted.



The observed barium binaries have periods anywhere from a couple hundred days up to tens of thousands of days. The short period binaries² are hard to explain with WRLOF only, while having orbital periods that are too *long* to be explained by a common envelope event. We should thus look for δ -values that produce these short period systems, which we find for δ -values as low as 0.2.

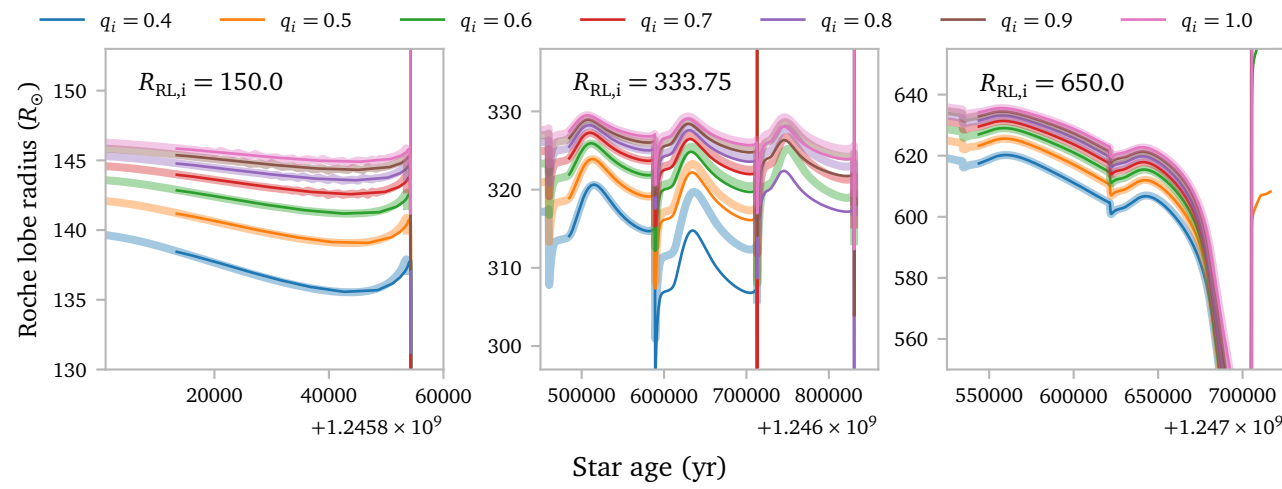
² Say less than 1000 days

b. Grid

In this section, I analyze the grid that I simulated with starting Roche lobe Radii from $150 R_{\odot}$ to $650 R_{\odot}$ in 10 logarithmically spaced steps and $7 q = M_d/M_d$ values from 0.4 to 1.

b.1 Validation

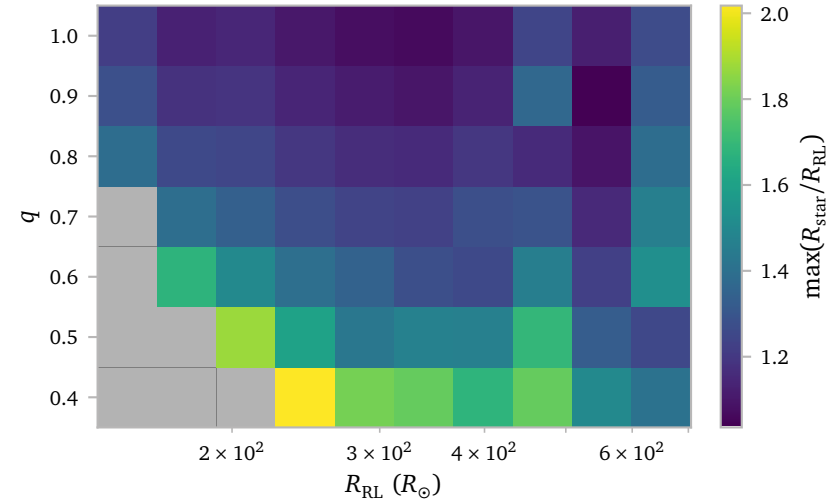
The first thing i want to do is make sure the models are actually starting where and how they are supposed too. To this end, I plot the `evolve.py` models from which the models are supposed to start, together with the MESA models for a couple of Roche lobe radii and the different q values. The plot below shows this.



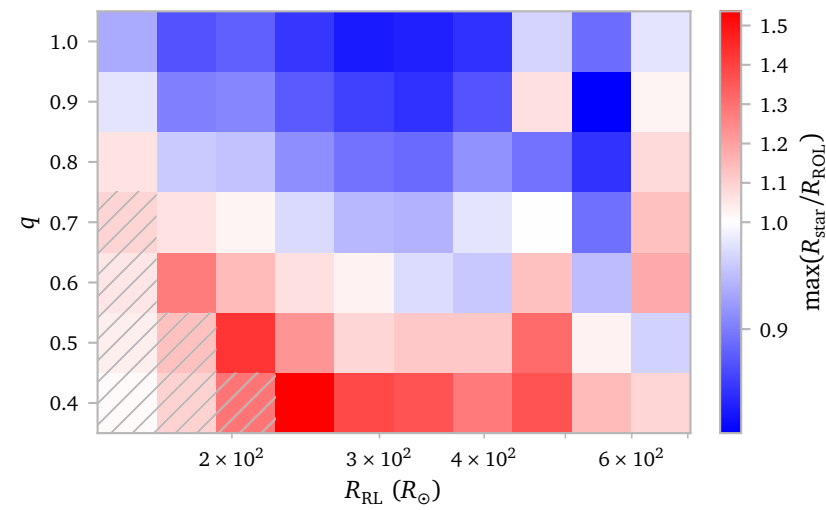
I think this looks good! The models all start where I expect them to start and they, at least initially, follow the predictions from `evolve.py` very closely.

b.2 General properties

I start with some general properties of the models displayed in a 2d grid.

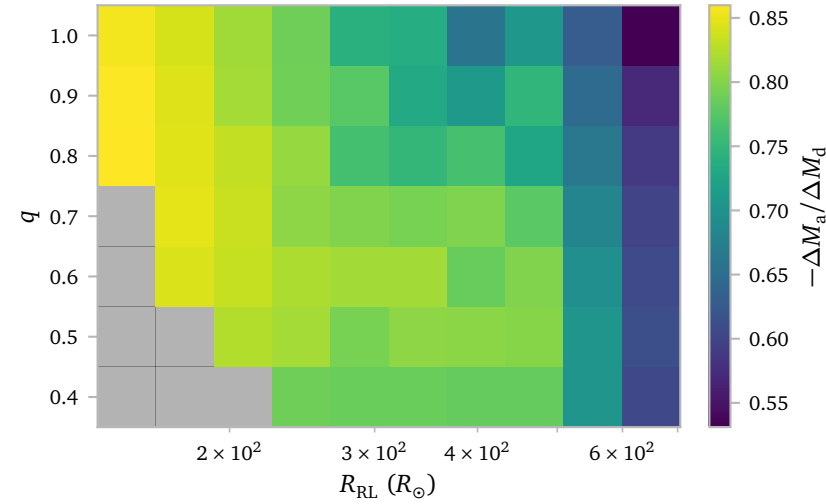


Here, I plot the maximum ratio of the star radius over the Roche lobe radius for every model. Initial q -values close to 1 seem to result in less extreme ratios. This makes sense, as for approximately conservative mass transfer, the orbit will reach its minimum separation when $q \approx 1$. For initial $q < 1$, this means the orbit will first shrink as the mass ratio approaches unity. This effect becomes less pronounced the closer the initial mass ratio is to unity. Once the winds become important, This relation becomes less clear. In the case of winds, the orbit will generally grow less for mass ratios close to unity than it will for mass ratios far from unity.

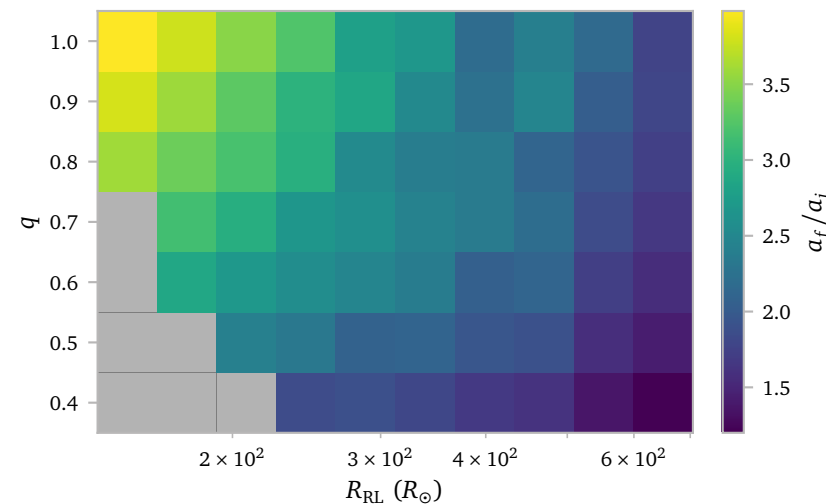


Similarly, we can plot the maximum ratio of the stellar radius over the volume equivalent outer Roche lobe radius using the analytical approximation from Temmink et al. (2023). The plot shows the same features but I use a divergent color map to show models where the ratio is greater and smaller than unity. Since the MESA models do not take outer lobe overflow into account, and the assumption of spherical symmetry can no longer be trusted when the stellar radius is larger than the outer Roche lobe, this is an important plot.

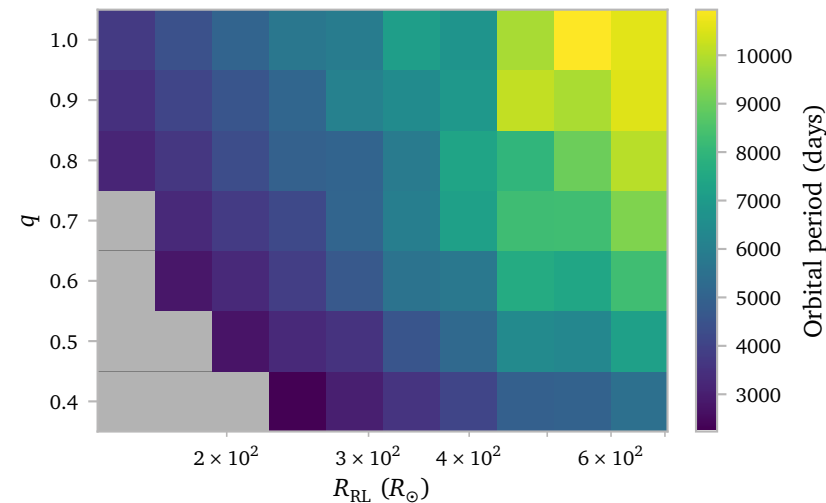
I think the third to last column has some surprising results. I investigate this in more detail in subsection b.3



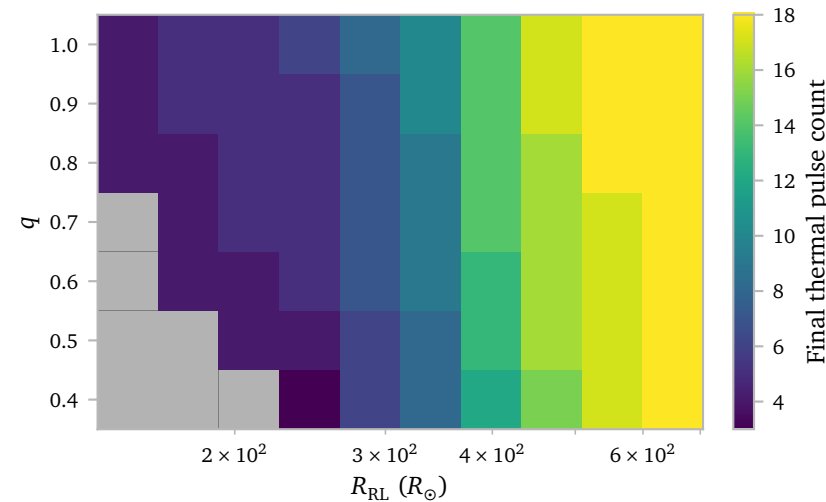
This plot shows the ratio of mass gained by the accretor and mass lost from the donor. A *higher* value means *more* of the mass that is lost from the donor successfully accretes on the accretor. The general trend of the plot seems to be: Larger initial Roche lobe radii lead to a smaller fraction of accreted mass. This makes sense since wind mass loss, which has a maximum accretion efficiency of about 30% becomes more important at larger radii.



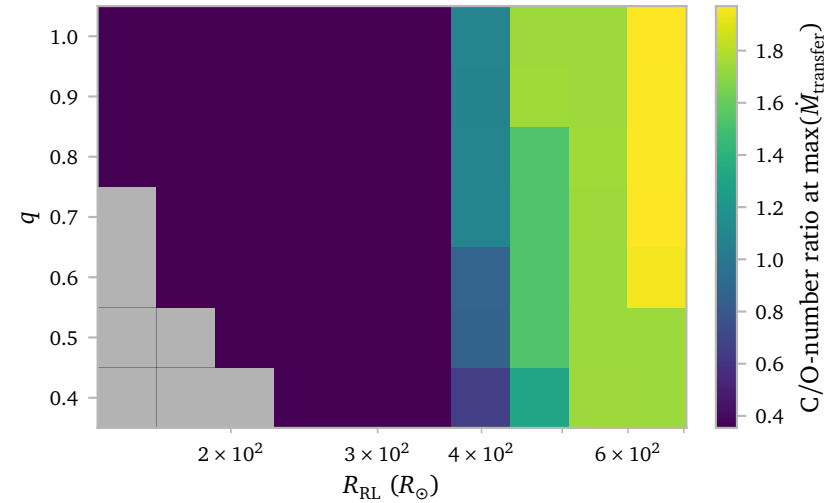
I think the results from this figure can be understood by combining the results section a. and the plot above. We see that the fraction of mass that is being accreted decreases with Roche lobe radius, and that the ratio of initial to final separation increases with q for conservative mass transfer. In general, a significant amount of mass is accreted, even when wind becomes important, but when this is the case, the relation between q and a_i/a_f becomes less pronounced.



This plot relates the amount of mass lost from the system, the plot above, and the initial periods. Systems with larger q tend to have larger initial separations. Systems with larger initial Roche lobe radii also tend to have larger initial separations. Combining this with mass lost due to winds, the largest periods occur for systems in the top right.



This the amount of thermal pulses the donor star experienced before becoming a post-AGB star. Perhaps the result is unsurprising. Stars starting with lower initial Roche lobe radii tend to lose their envelope quicker and thus experience less pulses. I guess the most interesting thing here is that stars with the same initial Roche lobe radius can experience a different number of thermal pulses depending on the mass of their companion.

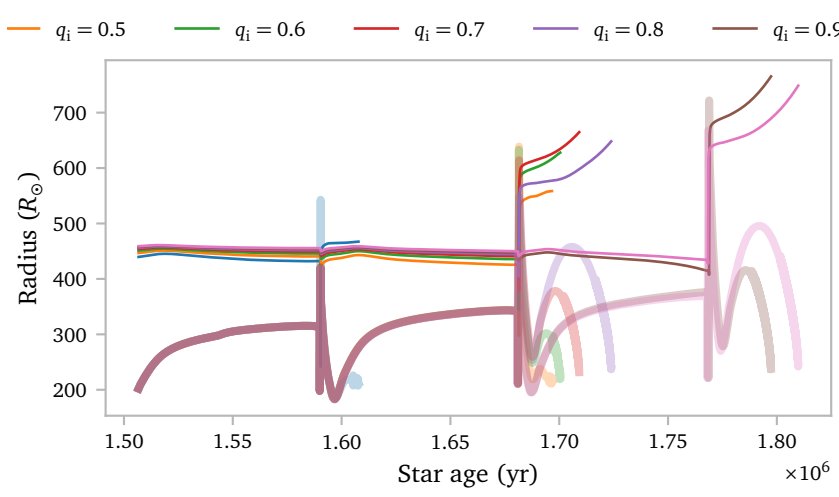


This figure shows the number ratio of C/O at the time of the maximum mass transfer rate. Since most of the mass is transferred during a single mass transfer episode during a thermal pulse, this is a good first approximation for the C/O-ratio of the material that gets accreted onto the accretor. It follows the plot above nicely, which is very much expected. There is quite a sudden jump though in the C/O-ratio. I *think* this is quite typical of TPAGB stars (at least $2 M_{\odot}$ TPAGB stars).

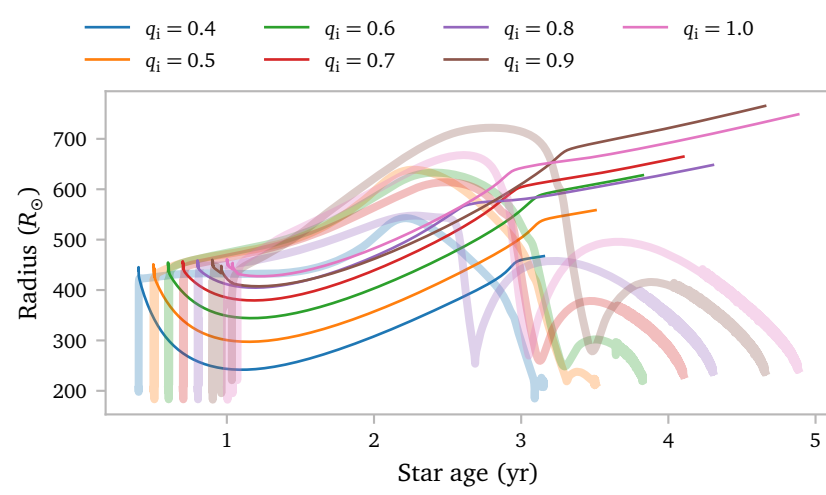
I think these plots, in particular the plot showing the C/O-ratio and the plot showing the final period, highlight that it is difficult to create the orbital properties of short period barium stars when assuming conservative RLOF mass transfer. Some mass probably has to be lost at the accretor to shrink the orbit sufficiently to achieve short period binaries.

b.3 $R_{RL} = 469.24 R_{\odot}$

Below I plot the time evolution of the third to last column of the grid.



The plot immediately shows that there is quite some difference between the pulse at which the most significant mass transfer episode starts. The peculiar plot from the section above showed a clear break in the trend for the maximum ratio of the star radius over the Roche lobe radius for the two largest initial q -values. We see that these systems transfer mass in a later pulse. I think it makes sense that these systems then experience a larger ratio during the next pulse since they reach it at an earlier point in the next pulse. The star is then able to expand more.



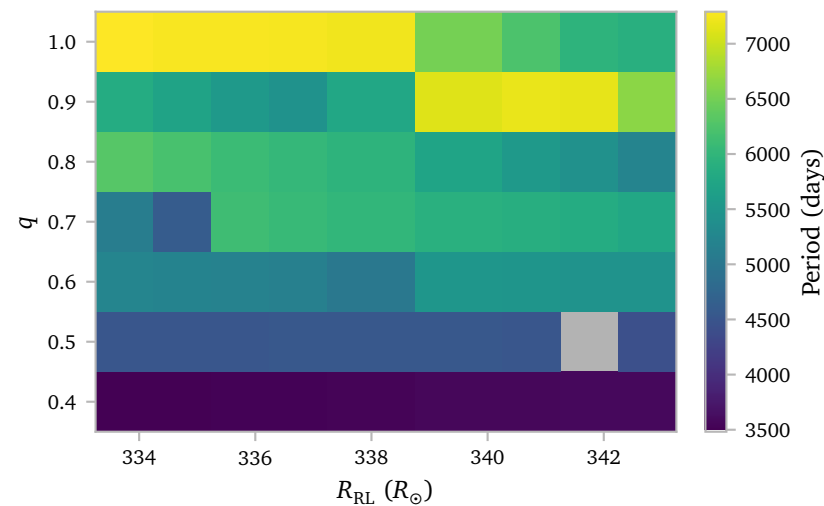
This plot more clearly shows that the radius of the stars becomes quite a bit larger for the two models that experience mass transfer during the later pulse. The stellar radius for the lowest mass ratio model is significantly smaller than that of the other models. This also makes sense as this model terminates during an earlier thermal pulse. The discontinuity thus is really just due to barely missing or barely hitting mass transfer during a pulse and if we sample a grid more finely, we should expect to see more of these discontinuities.

b.4 Fine Grid

In order to see how stable the results are to small variations in initial Roche lobe radius, I decided to simulate another grid. To test this, I decided to take a Roche lobe radius that I had already simulated and simulate 10 additional radii each differing by $1 R_{\odot}$.

I chose the reference radius as $R_{\text{RL}} \approx 338.75 R_{\odot}$. This is because for this radius, the binary systems with different q interacted during three different pulses, with some systems barely avoiding significant mass transfer. The systems also interact nicely during the middle to end of the TPAGB, so wind can have some effects while not overpowering the evolution.

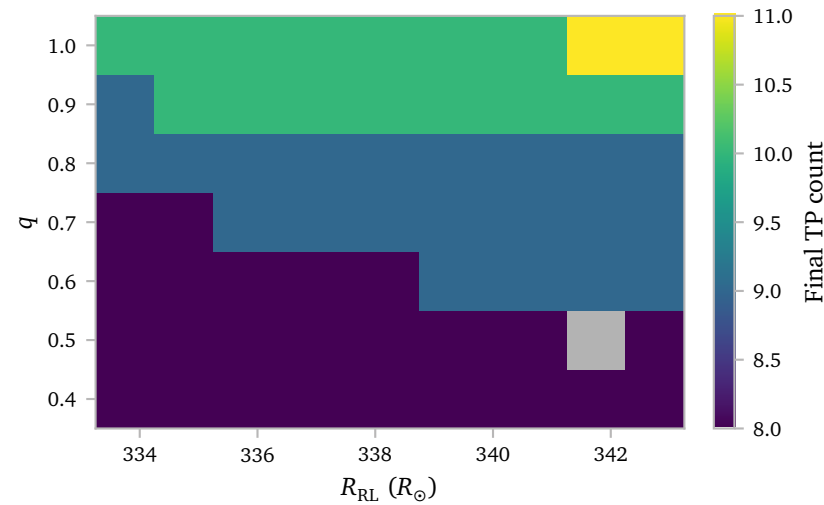
I was particularly interested in seeing how much the periods change with small perturbations in the initial Roche lobe radius. The plot below shows this.



In some cases, the differences are quite drastic. Nearly identical starting orbits can end with very different separations. I think it is important to try to understand where these differences come from.

Also, one of the models failed. It is interesting to look what caused this.

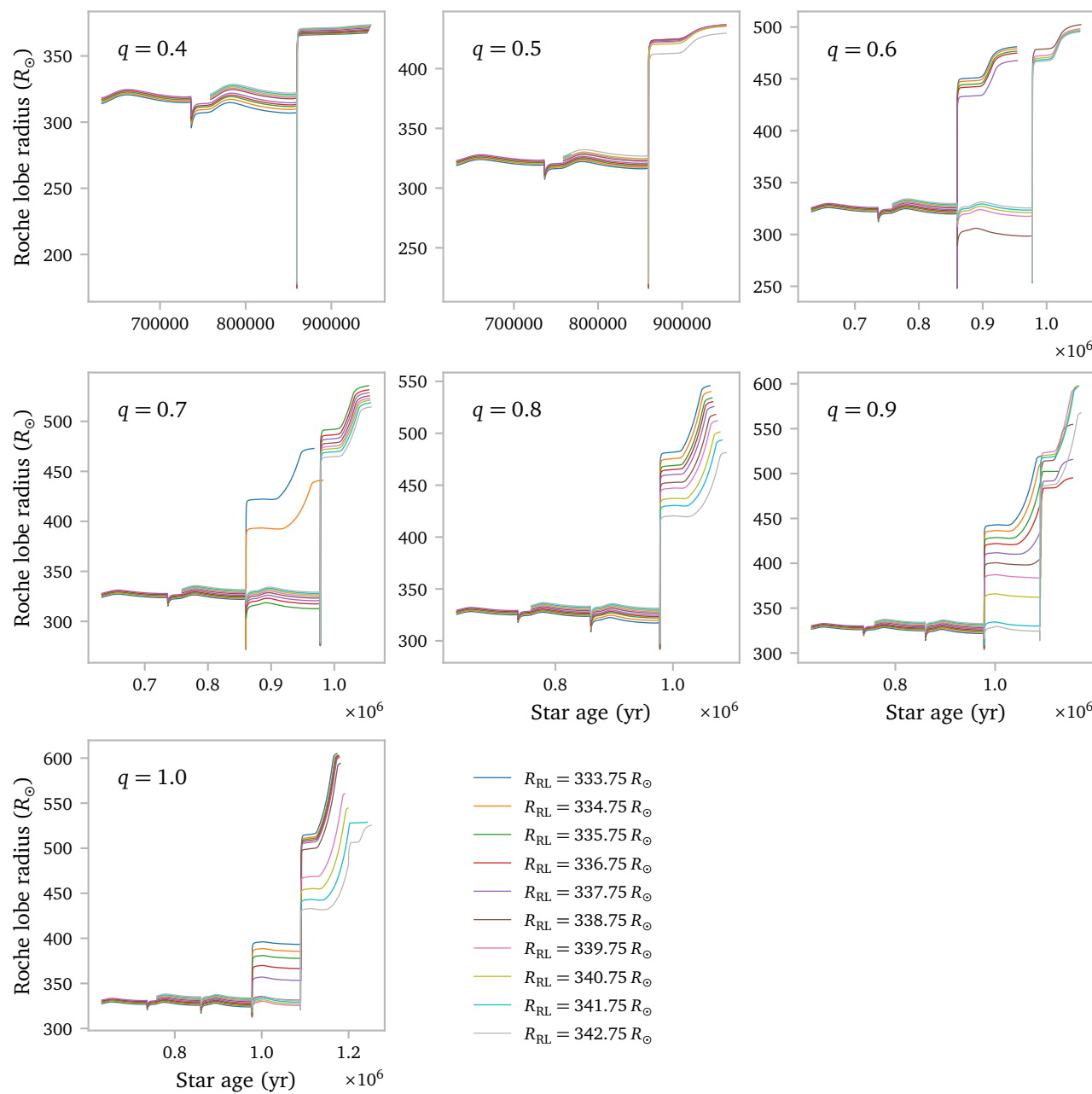
Presumably, this is due to interactions during a different thermal pulse. To check, we can look at the final TP count of the binary systems.



Although not a perfect match (for example, $q = 0.9$ shows nearly all systems experiencing the same number of pulses while clear differences in final periods exist) there does seem to be some relation between the thermal pulse count and the jumps in final period.

The plots below show the evolution of *all* models of this fine grid. They are split by q values. This makes it easier to compare. These are thus able to show the time evolution of the models of a single column in the plots above.

I start with a plot that shows the Roche lobe radius as a function of time.



The first two plots are not very interesting, all models seem to interact during the same pulse. The models stay order, meaning initially tighter orbits remain tighter after the evolution and overall, the differences between the models do not seem to diverge³. We see that the failing model in the $q = 0.5$ plot failed very early on in the evolution, without a clear reason.

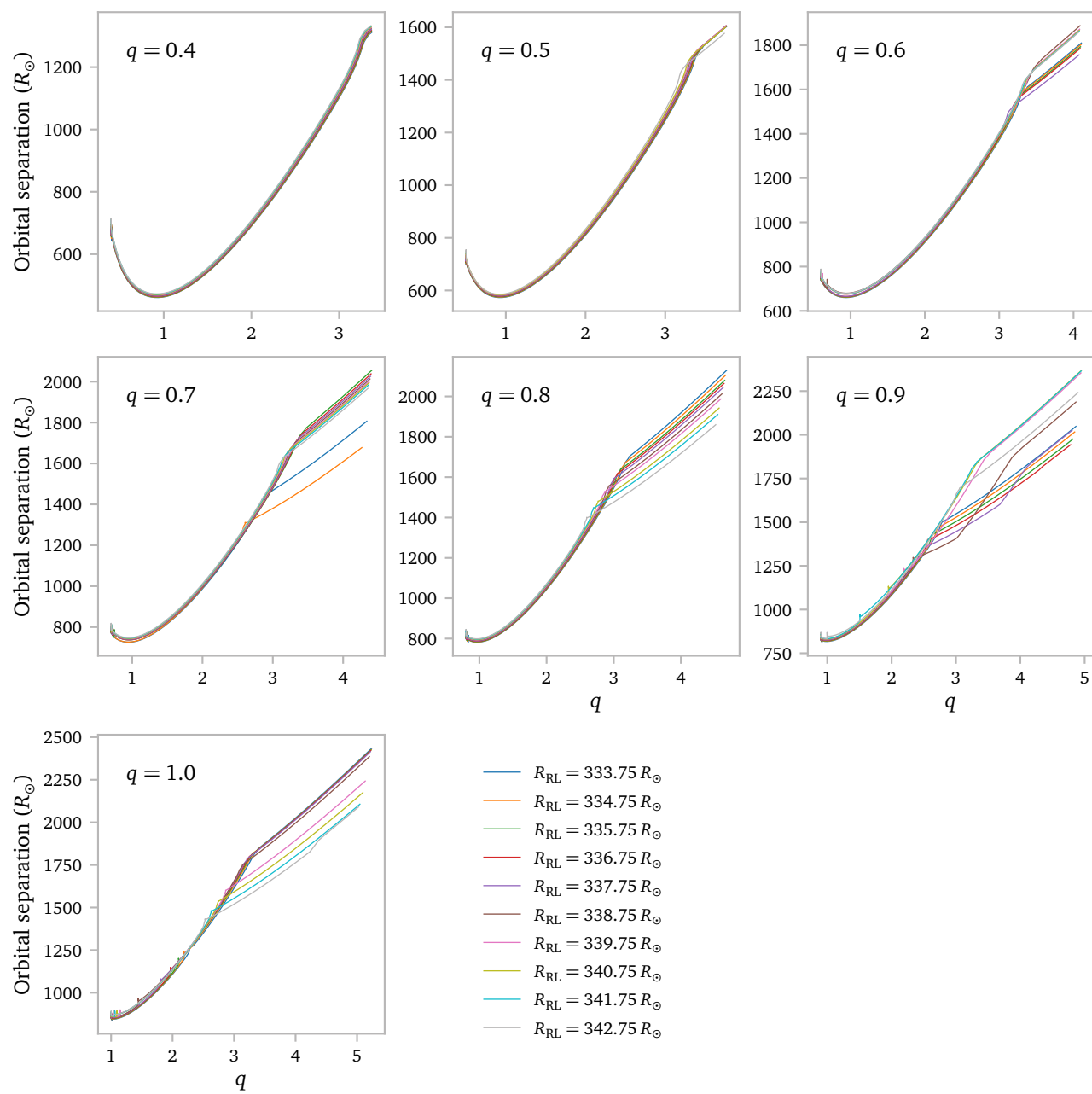
³ except for maybe the gray model in the $q = 0.5$ plot.

From $q = 0.6$ onwards⁴, we see some clear divergence in behaviour for some of the models. The divergence arises due to the tighter binaries just reaching the required conditions for significant RLOF while the more extended ones barely miss these conditions.

⁴ except for the $q = 0.8$ models

Another behaviour starts to appear from $q = 0.6$ onwards. That is the reversal of the Roche lobe radii after mass transfer. While the order remains intact for the low q models, after $q = 0.6$, the initially tighter binaries end up at larger radii and the initially looser binaries. I really do not know what causes this.

Below i plot the evolution of the binary separation as a function of q .

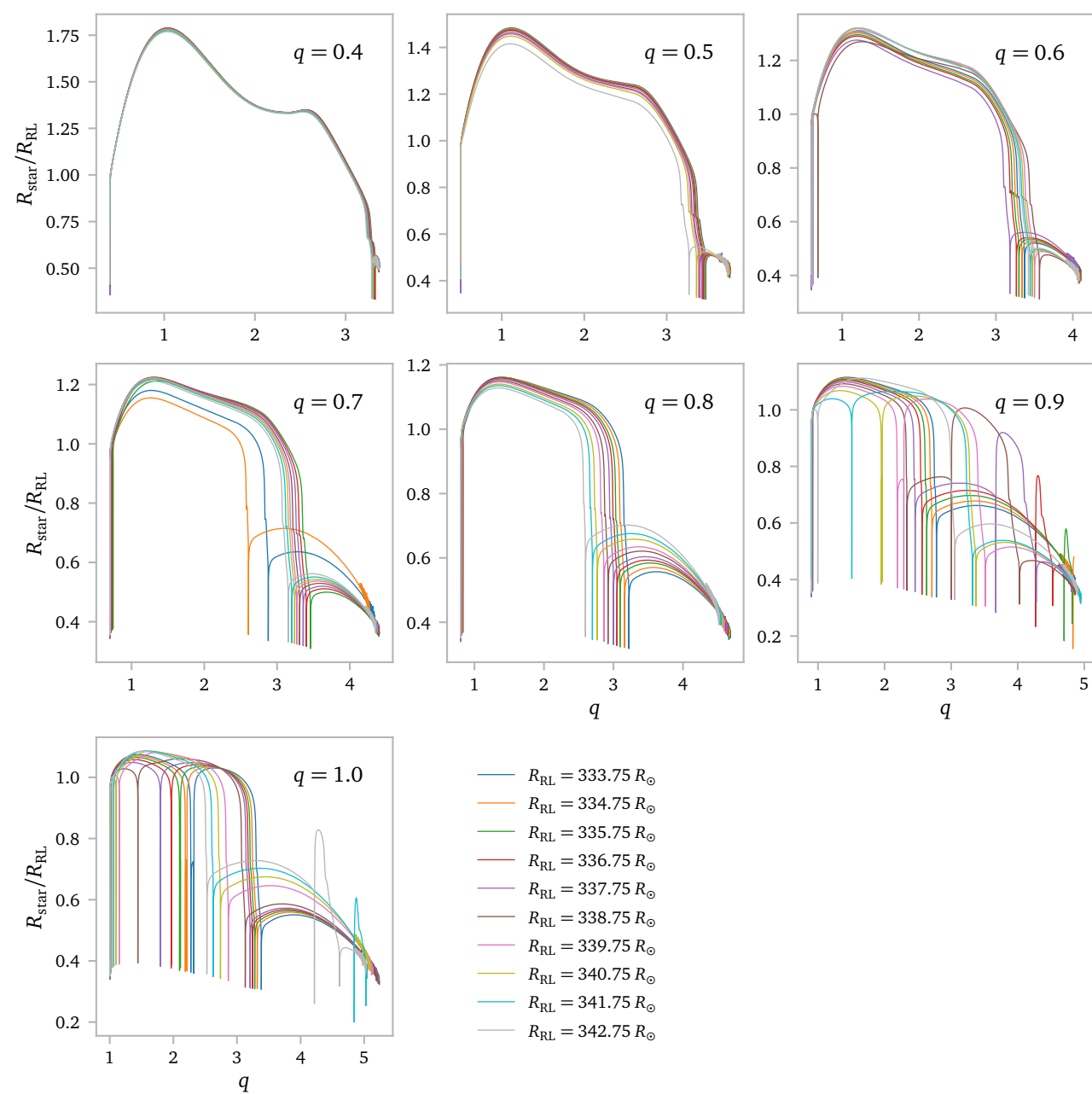


In this plot, it very easy to spot mass transfer due to winds⁵ and the regions of RLOF. It is clear that for greater q , winds become more important. This must be because the smaller q models are able to sustain RLOF for longer before their orbits become too wide.

⁵ Straighter lines

Actually, for these radii, the cut-off for RLOF seems to lie at about $q = 3$. I am not entirely sure why this is and it probably has to do with the ability of the star to expand in response to mass loss.

The plot below shows the ratio of the star radius over the Roche lobe radius again as a function of the mass ratio.

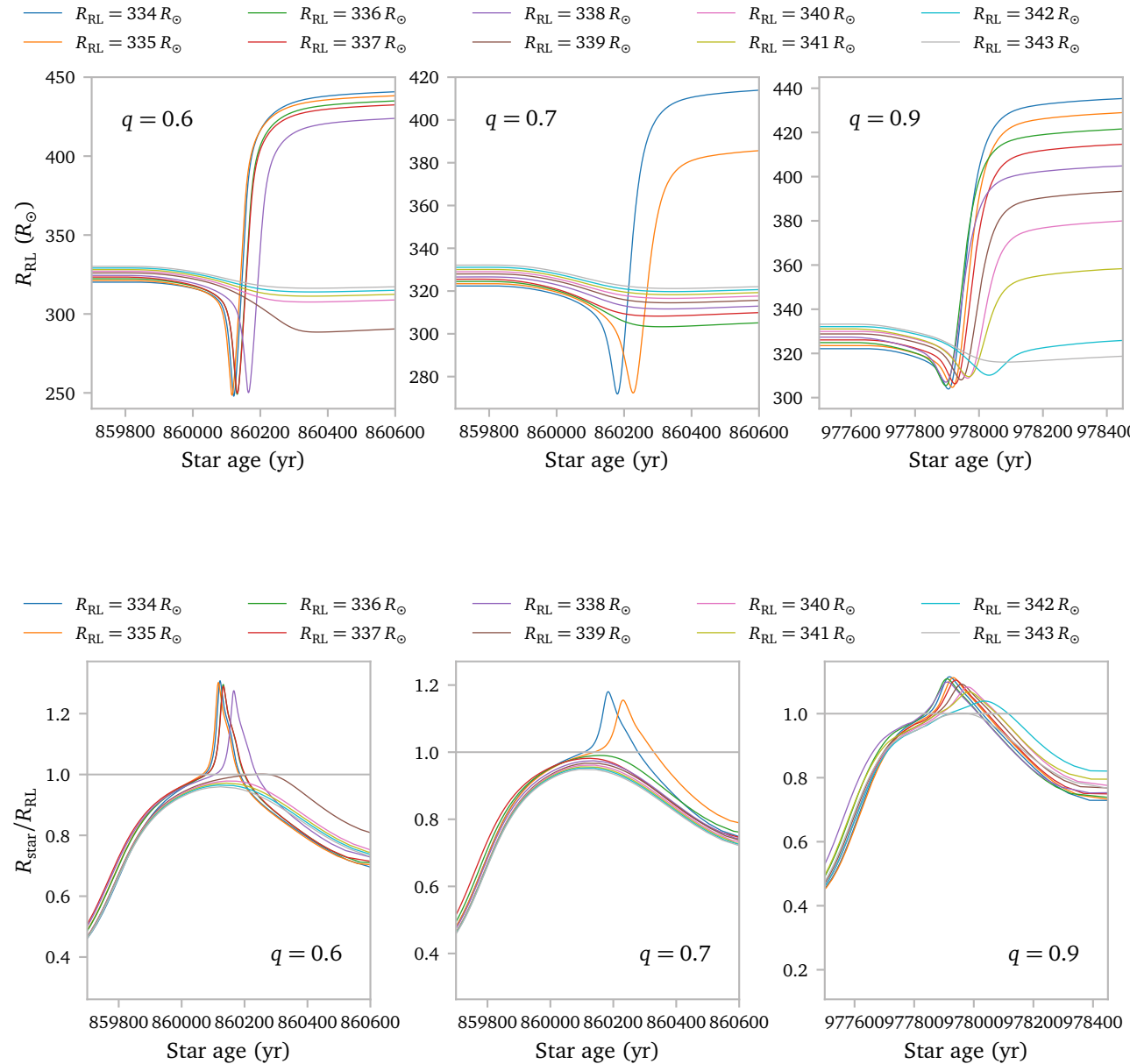


The same features are visible here. The last subplots get very messy, because the amount of mass transferred during the pulses is apparently very variable in these cases.

Lastly, we can zoom in a bit so hopefully see the separation between models that just barely start a RLOF mass transfer episodes and models that barely miss it. I look at the $q = 0.6, q = 0.7$ and $q = 0.9$ models because as seen in the figures above, there are clear separations visible here.

The upper panels show the Roche lobe radius, while the lower panels show the ratio. Unsurprisingly⁶, we see that models that fail to reach $R_{\text{star}}/R_{\text{RL}} > 1$ do not start their mass transfer episode. In this case, the Roche lobe radius shrinks in response to a narrow avoidance. Obviously, in response to the mass transfer episode, the Roche lobe radius increases significantly and a clear divide originates.

6 But still kind of cool to see!



Lastly, let's look at the model that failed to evolve. Looking at the terminal output, we see the following error message:

```
reduced varcontrol_target = 2.6000000000000100E-005
reduced varcontrol_target = 2.5000000000000099E-005
reduced varcontrol_target = 2.4000000000000099E-005
reduced varcontrol_target = 2.3000000000000098E-005
reduced varcontrol_target = 2.2000000000000098E-005
reduced varcontrol_target = 2.1000000000000097E-005
failed in do_saves_for_binaryphotos/b_x050
DATE: 2026-04-19
TIME: 00:01:51
finished
```

This is the code that is responsible for error message:

```
! $MESA_DIR/binary/private/binary_photos.f90
subroutine do_saves_for_binary(b, ierr)
  type(binary_info), pointer :: b
  integer, intent(out) :: ierr
  integer :: iounit, id
  character (len = strlen) :: str_photo, filename, iomsg, report_str

  call string_for_model_number('x', b% model_number, b% photo_digits, str_photo)

  filename = trim(trim(b% photo_directory) // '/' // str_photo)
  report_str = trim('save ' // filename)
  open(newunit = iounit, file = trim(filename), action = 'write', &
    status = 'replace', iostat = ierr, iomsg = iomsg, form = 'unformatted')
  if (ierr /= 0) then
    write(*, *) 'failed in do_saves_for_binary', trim(filename)
    return
  end if
  call binary_photo_write(b% binary_id, iounit)
  close(iounit)

  if (b% have_star_1) then
    filename = trim(trim(b% s1% photo_directory) // '/' // str_photo)
    call star_save_for_restart(b% s1% id, filename, ierr)
    report_str = trim(trim(report_str) // ', ' // filename)
  end if
  if (b% have_star_2) then
    filename = trim(trim(b% s2% photo_directory) // '/' // str_photo)
    call star_save_for_restart(b% s2% id, filename, ierr)
    report_str = trim(trim(report_str) // ', ' // filename)
  end if
  if (ierr /= 0) then
    write(*, *) 'failed in do_saves_for_binary'
    return
  end if

  write(*, *) trim(trim(report_str) // ' for model'), b% model_number
```

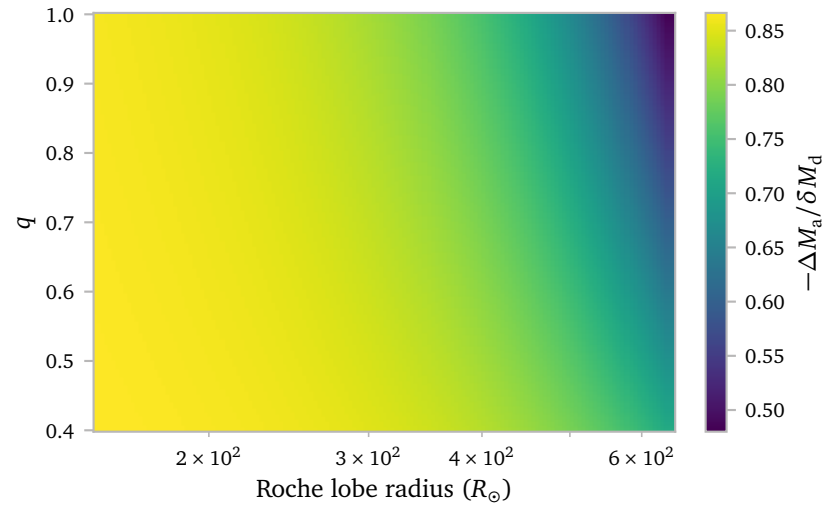
It terminates at the first `write(*,*)` call. I think it correctly gets the file-name. I am not sure why the model actually fails, especially because all other ones did not fail.

I was not able to reproduce the error locally on my desktop. The model ran without any trouble past the first few model save steps.

c. Choosing Wind and Accretion from the MESA models

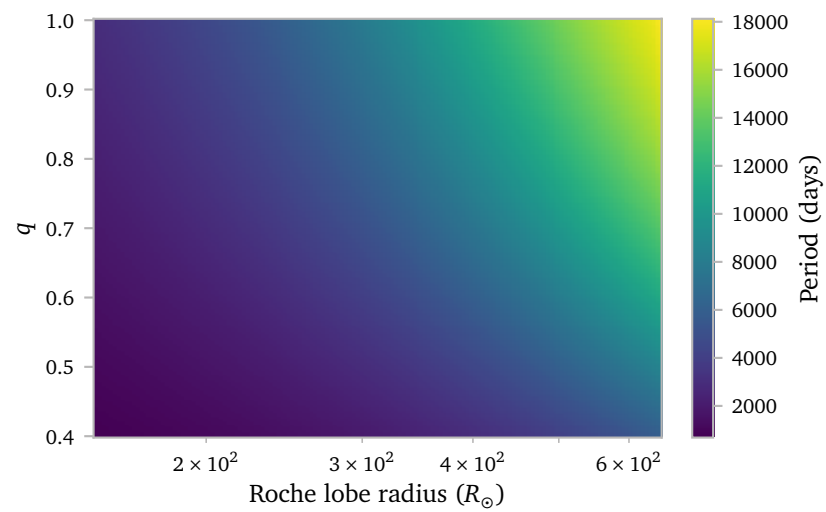
As a sort of first attempt to reproduce the orbital periods from MESA, I try to chose some informed values for α and β to hopefully find periods that are similar to the MESA models.

I start by trying to recreate the mass accretion efficiency plot from the MESA models



It is not perfect, but it matches the MESA models at least decently well. Here I model α based on some uninformed q and R_{RL} formulas. I then set $\beta = 0.3\alpha$ to account for the accretion due to Saladino et al. (2019) winds.

Next, I plot the orbital periods



The Soberman et al. (1997) prescription overestimates the periods. I think this is due to the differences between the Saladino et al. (2019) wind and the fast winds that Soberman et al. (1997) assumes for this star. We have previously already seen that fast winds tend to increase the orbit, while Saladino et al. (2018) winds can first make the orbit significantly shrink before undergoing mass transfer due to RLOF. After a mass ratio reversal, the orbit can increase.

References

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Soberman, G. E., Phinney, E. S., & van den Heuvel, E. P. J. (1997) *Stability criteria for mass transfer in binary stellar evolution..* *A&A***327**, 620.

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